

On-Policy Distillation on HyperPod — code to eval dashboard

Qwen3.5-9B teacher → Qwen3.5-2B student (LoRA r=128) · GSM8K · no SFT · one ml.g5.24xlarge · driven from one notebook

SETUP — ONCE

1

Code

Infra template,
training & eval code,
notebook

2

Provision

CloudFormation:
EKS, HyperPod, FSx,
S3

3

Configure

Code & weights to
S3, one ConfigMap

BASELINE — ONCE

4

Baseline

Serve the 9B; eval
9B · 2B · 0.8B
94.3 · 74.8 · 52.0%
→ pick the 2B

RUN — EACH EXPERIMENT

5

Distill

On-policy loop, 2B
LoRA student, no
warm-up

6

Evaluate

GSM8K: after vs. the
baseline and teacher

7

Dashboard

74.8% → 84.0%
teacher 94.3%
47% of the gap
closed

Inside step 5 — the on-policy loop

The student learns from the teacher's feedback on its own outputs

Student samples

64 prompts × 4 answers

Teacher scores

every token of those answers

Student updates

reverse KL · answer masked

repeat — the updated adapter is hot-swapped into the sampler every step

One node, four GPUs

Each GPU belongs to exactly one job

GPU 0

Teacher

GPU 1

Teacher

GPU 2

Student trainer

GPU 3

Student sampler + eval

/fsx FSx for Lustre — checkpoints, metrics, results

/s3 S3 — code and model weights